

1. N-QUEENS

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 20
```

```
int board[MAX];
```

```
bool isSafe(int row, int col) {  
    for (int i = 0; i < row; i++) {  
        if (board[i] == col ||  
            (i - row) == (board[i] - col) ||  
            (i - row) == (col - board[i])) {  
            return false;  
        }  
    }  
    return true;  
}
```

```
void printSolution(int n) {  
    printf("\nSolution:\n");  
  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            if (board[i] == j)  
                printf(" Q ");  
            else  
                printf(" . ");  
        }  
        printf("\n");  
    }  
}
```

```
bool solveNQueens(int row, int n) {  
    if (row == n) {  
        printSolution(n);  
        return true;  
    }  
  
    for (int col = 0; col < n; col++) {  
        if (isSafe(row, col)) {  
            board[row] = col;  
            solveNQueens(row + 1, n);  
        }  
    }  
}
```

```
    }

    return false;
}

int main() {
    int n;

    printf("Enter number of Queens: ");
    scanf("%d", &n);

    if (n < 1 || n > MAX) {
        printf("Invalid size!");
        return 0;
    }

    solveNQueens(0, n);

    return 0;
}
```



main.c

Output



Enter number of Queens: 4

Solution:

```
.  Q  .  .  
.  .  .  Q  
Q  .  .  .  
.  .  Q  .
```

Solution:

```
.  .  Q  .  
Q  .  .  .  
.  .  .  Q  
.  Q  .  .
```

=== Code Execution Successful ===

